

1. Introduction to Spring Framework

Spring is a lightweight, open-source framework for building Java applications. It provides infrastructure support for developing Java applications and helps manage objects and their dependencies using Inversion of Control (IoC) and Dependency Injection (DI).

2. What is Inversion of Control (IoC)?

IoC means giving control to the Spring container to manage the lifecycle and dependencies of objects, instead of creating objects manually using the 'new' keyword.

3. What is Dependency Injection (DI)?

DI is a design pattern where the dependencies of an object are provided externally. In Spring, this is done via XML or annotations.

4. Required Files for Basic Spring Program

1. Java Class (e.g., Student.java)
2. Spring Configuration XML file (applicationContext.xml)
3. Main class to test Spring (MainApp.java)

5. Maven Dependencies (pom.xml)

```
<dependencies>
<dependency>
<groupId>org.springframework</groupId>
<artifactId>spring-context</artifactId>
<version>5.3.30</version>
</dependency>
</dependencies>
```

6. Java Bean Class - Student.java

```
package com.example;
public class Student {
private String name;
public void setName(String name) { this.name = name; }
public void displayInfo() { System.out.println("Student name: " + name); }
}
}
```

7. Spring Configuration - applicationContext.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.springframework.org/schema/beans
http://www.springframework.org/schema/beans/spring-beans.xsd">
<bean id="studentBean" class="com.example.Student">
<property name="name" value="Noor Mustafa"/>
</bean>
</beans>
```

8. Main Class - MainApp.java

```
package com.example;
import org.springframework.context.ApplicationContext;
import
org.springframework.context.support.ClassPathXmlApplicationContext;
public class MainApp {
public static void main(String[] args) {
ApplicationContext context = new
ClassPathXmlApplicationContext("applicationContext.xml");
Student s = (Student) context.getBean("studentBean");
s.displayInfo();
}
}
```

9. Output of the Program

When you run MainApp.java, the output will be:
Student name: Noor Mustafa

10. Conclusion

You have learned how to create a basic Spring application using XML-based configuration. This includes IoC, DI, bean definition, and application context setup.